

Assignment 19: Product Inventory Management System (Salesforce Apex + Visualforce)

Step 1: Create Custom Object

Go to: **Setup** → **Object Manager** → **Create** → **Custom Object**

Create object:

- **Label:** Product Inventory
- **Plural Label:** Product Inventories
- **Object Name:** Product_Inventory
- **Record Name:** Product Name
- **Data Type:** Text

Click **Save**.

Step 2: Create Custom Fields

Inside Product Inventory object, create these fields:

1. Serial No

- Data Type: Number
- Field Label: Serial No
- Length: 10
- Decimal Places: 0
- API Name: `Serial_No__c`

2. Manufacture Date

- Data Type: Date
- API Name: `Manufacture_Date__c`

3. Expiry Date

- Data Type: Date
- API Name: `Expiry_Date__c`

Default Salesforce field:

- Product Name → `Name`
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Step 3: Create Apex Controller Class

Go to: **Setup** → **Apex Classes** → **New**

Class Name:

Paste code:

```
public class ProductController {

    public Product_Inventory__c product {get; set;}

    public ProductController() {
        product = new Product_Inventory__c();
    }

    public void saveProduct() {
        insert product;
        product = new Product_Inventory__c();
    }

    public List<Product_Inventory__c> getProducts() {
        return [
            SELECT Id, Name, Serial_No__c, Manufacture_Date__c,
            Expiry_Date__c
            FROM Product_Inventory__c
        ];
    }

    public PageReference deleteProduct() {
        String pid = ApexPages.currentPage().getParameters().get('pid');

        Product_Inventory__c p = [
            SELECT Id
            FROM Product_Inventory__c
            WHERE Id = :pid
            LIMIT 1
        ];

        delete p;
        return null;
    }
}
```

Click **Save**.

Step 4: Create Visualforce Page

Go to: **Setup** → **Visualforce Pages** → **New**

Page Name:

Paste code:

```
<apex:page controller="ProductController">
    <apex:form>
        <apex:pageBlock title="Product Inventory Management">

            <apex:pageBlockSection columns="1">
                <apex:inputField value="{!product.Name}" />
                <apex:inputField value="{!product.Serial_No__c}" />
                <apex:inputField value="{!product.Manufacture_Date__c}" />
                <apex:inputField value="{!product.Expiry_Date__c}" />
            </apex:pageBlockSection>

            <apex:commandButton value="Save Product" action="{!
saveProduct}" />

            <apex:pageBlockTable value="{!products}" var="p">
                <apex:column value="{!p.Name}" headerValue="Product Name" />
                <apex:column value="{!p.Serial_No__c}" headerValue="Serial
No" />
                <apex:column value="{!p.Manufacture_Date__c}"
headerValue="Manufacture Date" />
                <apex:column value="{!p.Expiry_Date__c}" headerValue="Expiry
Date" />

                <apex:column>
                    <apex:commandLink value="Delete"
                        action="{!deleteProduct}">
                        <apex:param name="pid" value="{!p.Id}" />
                    </apex:commandLink>
                </apex:column>
            </apex:pageBlockTable>

        </apex:pageBlock>
    </apex:form>
</apex:page>
```

Click **Save**.

Step 5: Run Application

Open the Visualforce page preview.

Enter:

- Product Name
- Serial No
- Manufacture Date
- Expiry Date

Click: **Save Product**

Products will display in the table.

To remove a product: Click: **Delete**

Features Implemented

- Add Product
 - View Products
 - Delete Product
 - Frontend using Visualforce
 - Backend using Apex
 - Custom object storage
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Viva Points

- `Product_Inventory__c` is a custom object.
- `Name` is Salesforce default record name field.
- Apex controller handles backend logic.
- Visualforce provides frontend UI.
- SOQL fetches product records.
- DML operations used: insert and delete.